

Global Trade Imbalances over the Past 70 Years: A Closer Look Using Penn World Table Data

DAVID CHI ZHANG

Summary

- The recent meeting of G20 finance ministers and central bank governors drew intense attention to global imbalances, which some analyses and media reports claim have reached an unsustainable historical peak.
- To assess this claim, this paper analyzes data from the Penn World Table (PWT), which covers GDP and merchandise trade from 1950 to 2023, and reaches three major findings:
 - Relative to global output, today's global merchandise trade imbalance is smaller than at its 2005–06 peak. In other words, we are not at an unprecedented level of global imbalance.
 - Since the 1980s, the US has generally had the world's largest deficit, while the position of largest surplus country has changed hands repeatedly; the US deficit relative to global GDP has followed a stable, autonomous cyclical pattern, and its characteristics and drivers should be a primary focus in analyses of global imbalances.
 - In recent years, China, the world's largest merchandise surplus country, has indeed contributed a steadily rising share of the aggregate global surplus. However, its share remains far below the historical extreme and more likely reflects a return toward the average pattern observed from 1955 to 1990.

Methodology

- Many articles and reports cite China's \$1.2 trillion trade surplus as critical evidence that current global imbalances are unprecedented and unsustainable.
- While this is an eye-catching number, several factors need to be considered in a more rigorous analysis:
 - If we are to discuss **global** imbalances, we should not examine any single country in isolation; we must first consider how aggregate surpluses and deficits across all countries have changed over time.
 - If we are to discuss the impact of imbalances on **the global economy**, we should focus not on the absolute dollar amounts of surpluses and deficits, but on their shares of global GDP.
 - If we are to identify the main contributors to imbalances, we need to see who has **persistently** run imbalances and measure the size of each country's contribution to the aggregate imbalance over time.
- This leads us to ask:
 - Is today's global trade imbalance at an unprecedented historical peak relative to the size of the global economy?
 - Which group exhibits more persistent imbalance dynamics: surplus countries or deficit countries?
 - Have the contributions of the countries with the largest imbalances to aggregate global imbalances reached a historical peak?

Methodology

- To answer these questions from a sufficiently long historical perspective, this paper analyzes data from the Penn World Table (PWT).
 - The PWT provides GDP and merchandise trade data covering 1950–2023.
- Calculations
 - First, the analysis computes each country's merchandise trade balance for every year and expresses it as a ratio to global GDP.
 - It then classifies each economy in each year as either surplus or deficit before computing aggregate global surpluses and deficits as shares of global GDP.

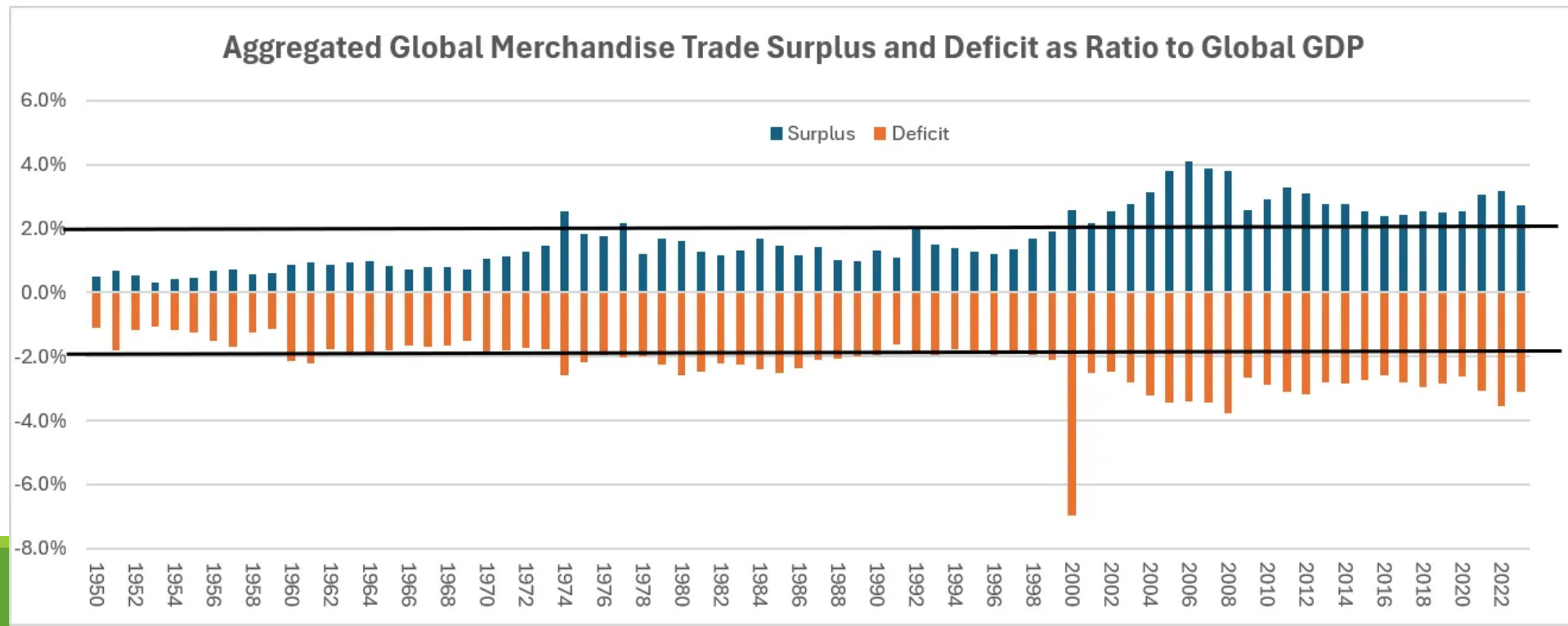
1. Historical pattern of aggregate global merchandise trade imbalances

- **Three phases stand out:**

- 1950–1970: the total surplus remained below 1% of global GDP, and the total deficit remained below 2%;
- 1970–2000: the total surplus fluctuated around 1.8% of global GDP, and the total deficit around 2%;
- 2000–present: both the total surplus and total deficit have remained above 2%, peaking at around 4% in 2005–06 before drifting lower amid fluctuations.

- **Finding 1: The current scale of global merchandise trade imbalances remains far below its 2005–06 peak**

- The claim that current global imbalances have reached an all-time extreme therefore does not hold up.



2. Largest surplus and deficit countries over time

- A stable leader among deficit countries (the US since 1983) and a rotating leader among surplus countries:
 - In the 1950s, the United States was the world's largest surplus country; in the 1960s and 1970s, Iran and Saudi Arabia held that position.

Country with the largest deficit	Country with the largest surplus	
Italy	United States	1950
Italy	United States	1951
Italy	United States	1952
Italy	Germany	1953
Italy	United States	1954
Italy	Iran (Islamic Republic of)	1955
Italy	United States	1956
Italy	United States	1957
Italy	Iran (Islamic Republic of)	1958
Italy	Iran (Islamic Republic of)	1959
Japan	Indonesia	1960
Japan	Iran (Islamic Republic of)	1961
Japan	Iran (Islamic Republic of)	1962
Japan	Iran (Islamic Republic of)	1963
Japan	United States	1964
Japan	Iran (Islamic Republic of)	1965
Japan	Iran (Islamic Republic of)	1966
Japan	Iran (Islamic Republic of)	1967
Japan	Iran (Islamic Republic of)	1968
Japan	Iran (Islamic Republic of)	1969
Norway	Iran (Islamic Republic of)	1970
United States	Saudi Arabia	1971
United States	Saudi Arabia	1972
Japan	Saudi Arabia	1973
Japan	Saudi Arabia	1974
Germany	Saudi Arabia	1975
Germany	Saudi Arabia	1976
United States	Iran (Islamic Republic of)	1977
United States	Saudi Arabia	1978
Germany	Saudi Arabia	1979
Germany	Saudi Arabia	1980
Germany	Saudi Arabia	1981
Germany	Saudi Arabia	1982

2. Largest surplus and deficit countries over time

- **A stable leader among deficit countries (the US since 1983) and a rotating leader among surplus countries:**
 - Since the early 1980s, the United States has been the largest deficit country in almost every year, while the position of largest surplus country shifted among Japan (1980s–1990s), Russia (around 2000), Saudi Arabia (2011–2012), and China (after 2012).

Country with the largest deficit	Country with the largest surplus	
United States	Japan	1983
United States	Japan	1984
United States	Japan	1985
United States	Japan	1986
United States	Japan	1987
United States	Japan	1988
United States	Japan	1989
United States	Japan	1990
United States	Japan	1991
United States	Russian Federation	1992
United States	Japan	1993
United States	Japan	1994
United States	Japan	1995
United States	Japan	1996
United States	Japan	1997
United States	Japan	1998
United States	Japan	1999
Bermuda	Japan	2000
United States	Russian Federation	2001
United States	Japan	2002
United States	Russian Federation	2003
United States	Russian Federation	2004
United States	Russian Federation	2005
United States	China	2006
United States	China	2007
United States	China	2008
United States	China	2009
United States	China	2010
United States	Saudi Arabia	2011
United States	Saudi Arabia	2012
United States	China	2013
United States	China	2014
United States	China	2015
United States	China	2016
United States	China	2017
United States	China	2018
United States	China	2019
United States	China	2020
United States	China	2021
United States	China	2022
United States	China	2023

2. Largest surplus and deficit countries over time

- **Surplus and deficit patterns in the US, Japan and Germany combined, and China:**
 - Since the 1980s, the US merchandise trade deficit has fluctuated cyclically around 1% of global GDP. Its current size is similar to its level around 1985 and below its 2005 peak.
 - Since the start of the 21st century, the US imbalance has shown no systematic correlation with the size of China's surplus: when the US imbalance peaked in 2005, China's surplus was less than one-third of the combined surplus of Japan and Germany and was rising only modestly; in 2005–2008, China's surplus expanded steadily while the US deficit narrowed rapidly; in 2015–2018, China's surplus declined steadily while the US deficit widened at the margin.
- **Finding 2: In short, the US imbalance appears to follow autonomous dynamics. Analyzing the characteristics and drivers of this stable pattern should be a primary focus when studying global imbalances.**



3. Contributions of the largest surplus and deficit countries over the past 70 years

- **Rising in recent years, but consistent with mean reversion**

- Since 2012, when China became the world's largest surplus country, the largest surplus country's share of the aggregate global surplus has indeed increased. However, viewed over the long run from 1950 to today, this rise looks more like mean reversion to a historical norm: the current share of about 35% is comparable to the average for 1955–1990. By contrast, the period from 2000 to 2010 appears to be the historical outlier.
- On the deficit side, the US contribution to the global deficit has edged up over the past decade. Although lower than in the first decade of the 21st century, the current contribution of about 35% is clearly elevated compared with 1950–1980, the period before the US became the persistent leader among deficit countries.

- **Finding 3: The largest surplus country's rising contribution in recent years looks more like mean reversion**

